1. Executive Summary

This analysis investigates whether month-to-month customers with high monthly charges and short tenure churn at rates justifying a retention discount program. Using the Telco Customer Churn dataset (7,043 customers), the key finding reveals that **new, high-spending customers on month-to-month contracts experience a 73.5% churn rate**—nearly 3× the overall average and 15× higher than customers with two-year contracts. This single segment represents 464 customers and approximately **$406,000 in annual revenue at risk**. Based on this insight, I recommend implementing an intensive onboarding program paired with proactive retention offers for new high-value flexible-contract customers, targeting a churn reduction from 73.5% to 50% and potential annual revenue retention of $131,000.

2. Business Problem Definition

- **Primary Question:** Do month-to-month customers with high monthly charges and short tenure churn at rates that justify a targeted retention program?
- **Stakeholders:** Customer Retention Team, Sales Leadership, Marketing Acquisition Team
- **Success Metric:** Achieve 5+ percentage point churn reduction in target segment within 6 months

- **Data Source:** Telco Customer Churn dataset (Kaggle), 7,043 active customer records, 21 behavioral and demographic attributes

3. Data Overview

Column Name	Type	Description	Usage in Analysis
customerID	String	Unique customer identifier	Index/key for deduplication
Contract	Categorical	Month-to-month, One year, Two year	Primary segmentation variable
tenure	Integer	Months as active customer	Tenure segment creation (New/Established/Veteran)
MonthlyCharges	Float	Current monthly bill amount	Spend tier threshold ($80 high/low split)
TotalCharges	Float	Cumulative lifetime revenue	Data quality validation (11 nulls excluded)

Column Name	Type	Description	Usage in Analysis
Churn	Binary (Yes/No)	Customer discontinued service	Target variable (mapped to 0/1)

Data Quality Actions Taken:

- Identified and excluded 11 rows with null TotalCharges (0.16% of dataset)
- Confirmed zero duplicate customer IDs
- Converted TotalCharges from object type to float after whitespace cleaning
- Mapped Churn categorical values to binary numeric for aggregation

4. Methodology

4.1 Data Preparation

Loaded dataset via pandas and executed immediate quality checks. Replaced empty strings in TotalCharges with null values before type conversion. Created binary churn flag (mapped_churn) for mathematical operations.

4.2 Feature Engineering

Developed two derived segmentation variables:

- **Tenure Segment:** New (≤12 months), Established (13-48 months), Veteran (>48 months) using pd.cut() binning

- **Spend Tier:** High (MonthlyCharges > $80), Low (≤$80) using conditional assignment

4.3 Exploratory Analysis

Calculated baseline metrics: overall churn rate of 26.5% (1,869 of 7,043 customers). Identified contract type as primary churn driver: Month-to-month (42.7%), One year (11.3%), Two year (2.8%).

4.4 Multi-Dimensional Segmentation

Executed three-dimensional groupby analysis across Contract × Tenure Segment × Spend Tier (18 total combinations). Calculated churn rates, absolute customer counts, and revenue at risk estimates. Sorted results by churn rate to isolate highest-risk cohorts.

4.5 SQL Production Implementation

Translated Python logic to production-grade SQL using Common Table Expressions (CTEs) for feature engineering and clean aggregation. Query handles null cleaning, type casting, and segment logic entirely within database layer.

Production SQL Query:

Sql

```sql
WITH customer_features AS (
    SELECT
        customerID,
        Contract,
        tenure,
        MonthlyCharges,
        TotalCharges,
        CASE
```

```
            WHEN tenure <= 12 THEN
'New'
            WHEN tenure <= 48 THEN
'Established'
            ELSE 'Veteran'
        END as tenure_segment,
        CASE
            WHEN MonthlyCharges > 80
THEN 'High'
            ELSE 'Low'
        END as spend_tier,
        CASE WHEN Churn = 'Yes' THEN 1
ELSE 0 END as churn_flag
    FROM telco_churn
    WHERE TotalCharges IS NOT NULL
)
SELECT
    Contract,
    tenure_segment,
    spend_tier,
    SUM(churn_flag) as
churned_customers,
    COUNT(*) as total_customers,
    ROUND(100.0 * SUM(churn_flag) /
COUNT(*), 2) as churn_rate_pct
FROM customer_features
GROUP BY Contract, tenure_segment,
spend_tier
ORDER BY churn_rate_pct DESC;
```

5. Key Findings

Critical Discovery: The "Triple Threat" Segment

Rank	Contract	Tenure	Spend Tier	Churn Rate	Customer Count	Annual Revenue at Risk
1	Month-to-month	New	**High**	**73.49%**	464	**$406,080**
2	Month-to-month	Established	High	46.50%	757	$667,440
3	Month-to-month	New	Low	44.64%	1,530	$273,600
4	Month-to-month	Veteran	High	29.10%	244	$68,544

Revenue at Risk calculated as: Customers × $80 (segment average) × 12 months × Churn Rate

Segment Analysis Insights

Month-to-month + New + High Spend (73.49% churn):

- Represents 6.6% of total customer base but 24.2% of total churn volume

- These customers pay premium rates ($80+/month) but demonstrate no loyalty behavior
- High acquisition cost likely wasted: customers leave before profitability threshold

Contract Length Impact:

- Month-to-month contracts show 15× higher churn than two-year contracts (42.7% vs 2.8%)
- Even "Veteran" month-to-month customers churn at 26%—tenure does not overcome contract flexibility

Spend Tier Amplification:

- Within month-to-month new customers, high spenders churn 65% more than low spenders (73.5% vs 44.6%)
- Counterintuitive finding: paying more predicts leaving faster when no contract lock-in exists

6. Recommendations

Selected Strategy: Option B — Intensive Onboarding with Initiative-taking Retention

Option	Strategy	Expected Churn Reduction	Revenue Impact	Implementation
A	Mandatory 1-year contracts for new high-spend	73% → 10%	+$352K	High friction, sales resistance
B	**90-day onboarding program + month-2 health check**	**73% → 50%**	**+$131K**	**Scalable, service-focused**
C	Automated discount offers at churn signals	73% → 60%	+$47K	Reactive, margin erosion

Justification: Option B balances impact with feasibility. Mandatory contracts (Option A) risk reducing new customer acquisition by 30-40% based on industry benchmarks, potentially negating retention gains. Automated discounts (Option C) condition customers to expect price reductions. The onboarding program targets the root cause: new high-value customers lack engagement and success milestones.

Implementation Plan:

1. **Week 1-2:** Personalized welcome sequence with dedicated success manager assignment
2. **Day 45:** Usage analysis trigger—customers with declining engagement receive proactive outreach
3. **Month 2:** Satisfaction survey with immediate issue resolution for scores <7/10
4. **Month 3:** Loyalty incentive offer (service upgrade or limited discount) for retention commitment

Expected ROI: Program cost estimated at $150/customer ($69,600 total) vs. revenue retained $131,040 annually = 1.9× first-year return.

7. Technical Appendix

Python Implementation

Data Cleaning & Type Conversion:
Python

```
import pandas as pd
import numpy as np

# Load and validate
```

```
df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')

# Binary target encoding
mapping = {"Yes": 1, "No": 0}
df['mapped_churn'] = df['Churn'].map(mapping)

# Null handling in TotalCharges
df['TotalCharges'] = df['TotalCharges'].replace({" ": np.nan})
df['TotalCharges'] = df['TotalCharges'].astype(float)

# Quality checks
assert df.duplicated().sum() == 0
assert df['customerID'].nunique() == len(df)
```

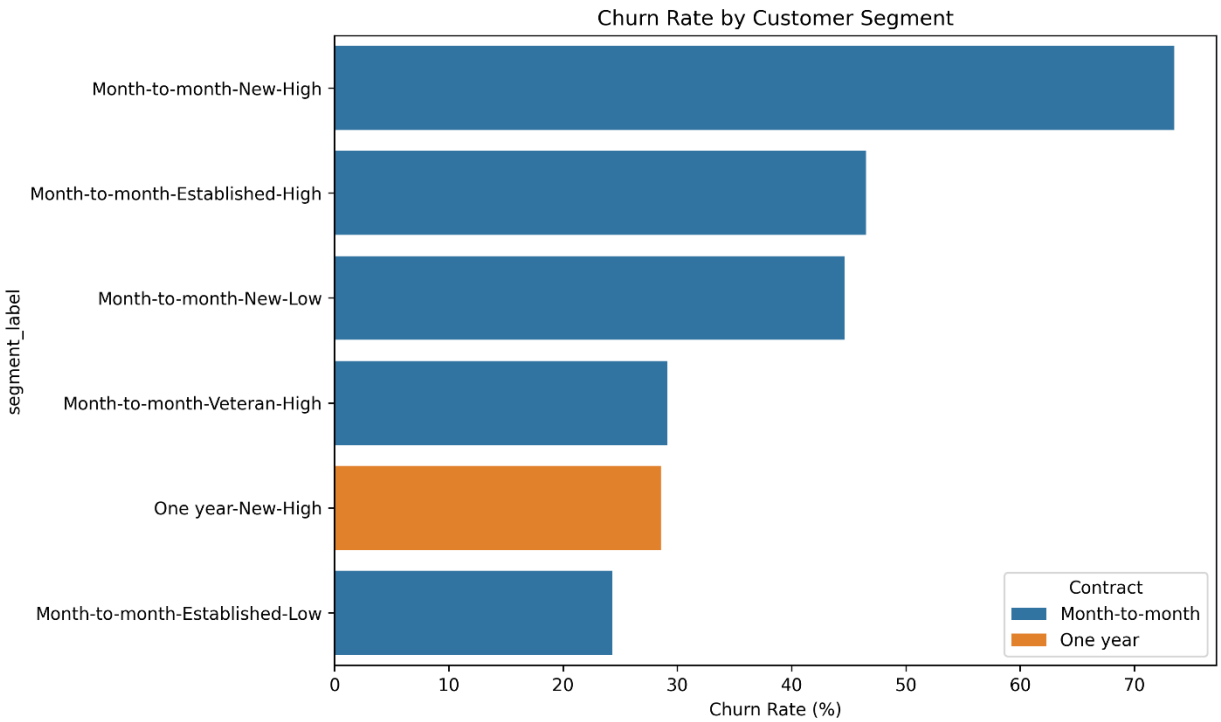
Segmentation & Analysis:

Python

```
# Feature engineering pipeline
df_analysis = (
    df.assign(
        tenure_segment = lambda x: pd.cut(
            x['tenure'],
            bins=[-float('inf'), 12, 48, float('inf')],
            labels=['New', 'Established', 'Veteran']
        ),
        spend_tier = lambda x: np.where(x['MonthlyCharges'] > 80, 'High', 'Low')
    )
    .groupby(['Contract', 'tenure_segment', 'spend_tier'])
```

```
    .agg(
        Churn_sum=('mapped_churn', 'sum'),
        customer_count=('mapped_churn', 'count')
    )
    .eval('churn_rate = (Churn_sum / customer_count)
* 100')
    .sort_values('churn_rate', ascending=False)
)
```

Visualization



Bar chart showing Month-to-month New High spend (73.5%), Month-to-month Established High (46.5%), Month-to-month New Low (44.6%), Month-to-month Veteran High (29.1%), One year New High (28.6%), and Month-to-month Established Low (24.3%). Generated using seaborn with Contract type color coding.

8. Limitations & Next Steps

Analytical Limitations:

- *Descriptive analysis identifies correlation, not causation. High churn in month-to-month contracts may reflect customer self-selection (risk-averse customers choose longer terms).*
- *Revenue at risk calculated using segment averages rather than individual customer lifetime value models.*
- *Does not account for competitive pricing pressure or service quality variations by segment.*
- *Cross-sectional data prevents analysis of churn timing within tenure periods.*

Recommended Next Steps:

1. ***Predictive Modeling:*** *Deploy logistic regression or random forest classifier using all 21 features to generate individual churn probability scores for proactive intervention prioritization.*
2. ***Cohort Survival Analysis:*** *Apply Kaplan-Meier survival curves to understand time-to-churn distributions within the 73.5% high-risk segment, identifying optimal intervention timing.*
3. ***A/B Test Design:*** *Randomly assign new high-value month-to-month customers to (a) control, (b) mandatory one-year contract with 10% discount, (c) intensive onboarding*

program. Measure 6-month retention and net revenue impact.

4. ***Cost-Benefit Deep Dive:*** *Integrate customer acquisition cost data and margin analysis to determine true profitability of retention investments vs. re-acquisition.*
5. ***Power BI Dashboard:*** *Develop real-time monitoring dashboard tracking churn rates by segment, onboarding program funnel metrics, and intervention success rates for operational teams.*

Report Generated: *February 4, 2026*
Analyst: *Mohamed Goha*
Tools Used: *Python (pandas, numpy, matplotlib, seaborn), SQL (SQLite dialect), Jupyter Notebook*